EXERCISE – 6.1

CORRELATION ANALYSIS

1. **Crimes** The number of murders and robberies per 100,000 population for a random selection of states is shown. Is there a linear relationship between the variables?

Murders	2.4	2.7	5.6	2.6	2.1	3.3	6.6	5.7
Robberies	25.3	14.3	151.6	91.1	80	49	173	95.8

(a) Construct Scatter Plot for the data and interpret.
(b) Find Correlation Coefficient and interpret the results.
(c) Find Coefficient of Determination also interpret the results.
(d) Test the significance of correlation coefficient using statistical inference.
$(Ans: r = 0.804,\ R^2 = 0.65)$

2. **State Debt and Per Capita Tax** An economics student wishes to see if there is a relationship between the amount of state debt per capita and the amount of tax per capita at the state level. Based on the following data, can she or he conclude that per capita state debt and per capita state taxes are related?

Per capita debt *x*	1924	907	1445	1608	661
Per capita tax *y*	1685	1838	1734	1842	1317

(a) Construct Scatter Plot for the data and interpret.
(b) Find Correlation Coefficient and interpret the results.
(c) Find Coefficient of Determination also interpret the results.
(d) Test the significance of correlation coefficient using statistical inference.
$(Ans: r = 0.518,\ R^2 = 0.27)$

3. **Faculty and Students** The number of faculty and the number of students are shown for a random selection of small colleges. Is there a significant relationship between the two variables? Switch *x* and *y* and repeat the process. Which do you think is really the independent variable?

Faculty	99	110	113	116	138	174	220
Students	1353	1290	1091	1213	1384	1283	2075

(a) Construct Scatter Plot for the data and interpret.
(b) Find Correlation Coefficient and interpret the results.
(c) Find Coefficient of Determination also interpret the results.
(d) Test the significance of correlation coefficient using statistical inference.
$(Ans: r = 0.812,\ R^2 = 0.66)$

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4. Class Size and Grades School administrators wondered whether class size and grade achievement (in percent) were related. A random sample of classes revealed the following data. Are the variables linearly related?

No. of students	15	10	8	20	18	6
Avg. grade (%)	85	90	82	80	84	92

(a) Construct Scatter Plot for the data and interpret.
(b) Find Correlation Coefficient and interpret the results.
(c) Find Coefficient of Determination also interpret the results.
(d) Test the significance of correlation coefficient using statistical inference.
$(Ans: r = -0.673,\ R^2 = 0.45)$

5. Alumni Contributions The director of an alumni association for a small college wants to determine whether there is any type of relationship between the amount of an alumnus's contribution (in dollars) and the number of years the alumnus has been out of school. The data follow

Years x	1	5	3	10	7	6
Contribution y	500	100	300	50	75	80

(a) Construct Scatter Plot for the data and interpret.
(b) Find Correlation Coefficient and interpret the results.
(c) Find Coefficient of Determination also interpret the results.
(d) Test the significance of correlation coefficient using statistical inference.
$(Ans: r = -0.883,\ R^2 = 0.78)$

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